The Volume Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: AREA, PRISM, PYRAMID and MODEL (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (AREA)
- ☐ **A2** (AREA)
- ☐ **A3** (PRISM)
- ☐ **A4** (PYRAMID)
- ☐ **A5** (PYRAMID)
- ☐ **A6** (MODEL)

Entrées

10 pts each · choose 4

- ☐ **E1** (AREA)
- ☐ **E2** (PRISM)
- ☐ **E3** (PRISM)
- ☐ **E4** (PYRAMID)
- ☐ **E5** (MODEL)
- ☐ **E6** (MODEL)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (AREA)
- ☐ **S2** (PRISM)
- ☐ **S3** (PYRAMID)
- ☐ **S4** (MODEL)

My skill check (circle each skill you have covered): AREA · PRISM · PYRAMID · MODEL

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 AREA ☐ My pick · 5 pts

Find the area of a trapezoid with parallel sides 8 cm and 14 cm and height 5 cm.

A2 AREA ☐ My pick · 5 pts

An L-shaped patio is a 10 m by 6 m rectangle with a 4 m by 3 m rectangular corner removed. Find its area.

A3 PRISM ☐ My pick · 5 pts

A rectangular prism is 5 in by 4 in by 3 in. Find its volume and its surface area.

A4 PYRAMID ☐ My pick · 5 pts

Find the volume of a cone with radius 3 cm and height 7 cm. Leave your answer in terms of π.

A5 PYRAMID ☐ My pick · 5 pts

A sphere has radius 6 in. Find its volume and its surface area, in terms of π.

A6 MODEL ☐ My pick · 5 pts

A room is 4 yd by 5 yd. Find its area in square feet. (1 yd = 3 ft)

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 AREA ☐ My pick · 10 pts

A window is a rectangle 4 ft wide and 6 ft tall, topped by a semicircle whose diameter is the rectangle’s width. Find the total area in terms of π, then to the nearest tenth of a square foot.

E2 PRISM ☐ My pick · 10 pts

A triangular prism has a right-triangle base with legs 6 cm and 8 cm and hypotenuse 10 cm. The prism is 12 cm long. Find its volume and its total surface area.

E3 PRISM ☐ My pick · 10 pts

A cylinder has volume 150π cm³ and height 6 cm. Find its radius, then its total surface area, in terms of π.

E4 PYRAMID ☐ My pick · 10 pts

A square pyramid has base edge 10 cm and slant height 13 cm. **(a)** Find its surface area. **(b)** Use the height, half the base, and the slant height to find the pyramid’s height, then its volume.

E5 MODEL ☐ My pick · 10 pts

A box is 2 ft by 3 ft by 1 ft. Kim says: “Its volume is 6 ft³, which is $6 \times 12 = 72$ in³.” Find Kim’s mistake and give the correct volume in cubic inches.

E6 MODEL ☐ My pick · 10 pts

A tank is 50 cm long, 30 cm wide and 40 cm tall, filled with water to 5 cm below the top. (1 L = 1000 cm³) **(a)** How many liters of water are in it? **(b)** If 12 L is removed, how far does the water level drop?

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 AREA ☐ My pick · 20 pts

A park is a 40 m by 30 m rectangle. A 12 m by 12 m square playground and a semicircular pond with diameter 20 m lie inside it, not overlapping. **(a)** Find the grass area in terms of π, then to the nearest m². **(b)** A bag of seed covers 150 m². How many bags are needed? **(c)** Rosa says doubling the pond’s diameter doubles its area. Is she right? Explain.

S2 PRISM ☐ My pick · 20 pts

Can A has radius 4 cm and height 9 cm. Can B has radius 6 cm and holds the same volume. **(a)** Find Can A’s volume. **(b)** Find Can B’s height. **(c)** Find each can’s label area (side only). **(d)** Which can uses less label material, and by how much? Use π in answers.

S3 PYRAMID ☐ My pick · 20 pts

An ice-cream cone has radius 3 cm and height 12 cm. A hemisphere of ice cream with the same radius sits on top. **(a)** Find the cone’s volume. **(b)** Find the hemisphere’s volume. **(c)** The cone is hollow. If all the ice cream melts into it, will it overflow? Explain with numbers. **(d)** Find the cone’s slant height in simplest radical form.

S4 MODEL ☐ My pick · 20 pts

A pool’s inside is a rectangular prism 12 m long, 6 m wide and 1.5 m deep. **(a)** Find its volume in m³ and in liters (1 m³ = 1000 L). **(b)** A pump adds 450 L per minute. How many hours to fill it? **(c)** The floor and four walls are tiled with 0.5 m by 0.5 m tiles. How many tiles are needed (no waste)?